

Prevention of diabetes and reduction in major cardiovascular events in studies of subjects with prediabetes: meta-analysis of randomised controlled...



BACKGROUND: Impaired glucose tolerance (IGT) and impaired fasting glucose (IFG) are pre-diabetic states, treatment of which may prevent or delay the onset of overt diabetes and thus potentially reduce major cardiovascular (CV) events. We therefore sought to determine whether interventions (including diet, exercise and pharmacological therapy), altered all-cause and cardiovascular related mortality in such subjects. **METHODS:** We performed a meta-analysis of prospective, randomised controlled trials (RCTs) that were identified in the medical literature and databases. Trials were eligible for inclusion if they reported all-cause mortality rates (at a minimum), recruited approximately 100 patients and had a minimum follow-up of one year. Interventions were divided into pharmacological and non-pharmacological. **RESULTS:** Ten RCTs that enrolled 23,152 patients met the above entry criteria. Trials ran for an average of 3.75 years. Diabetes was delayed or prevented by these interventions vs control (risk ratio 0.83, 95%CI 0.80-0.86). Non-drug approaches (n = 3495) were superior to drug-based approaches (n = 20,872) in diabetes prevention (0.52, 0.46-0.58 vs 0.70, 0.58-0.85, $P < 0.05$). There was no difference in risk of all-cause mortality in the intervention versus control group (0.96, 0.84-1.10) and no difference in CV death (1.04, 0.61-1.78). There was a non-significant trend towards reduction in fatal and non-fatal myocardial infarction (0.59, 0.23-1.50). Fatal and non-fatal stroke was borderline reduced (0.76, 0.58-0.99) with intervention versus control. **CONCLUSIONS:** Despite interventions being mostly successful in retarding progression to overt diabetes, this did not result in reductions in all-cause or cardiovascular mortality, or myocardial infarction, with the possible exception of stroke.